



A 2023 socio-ecological imagination: What if a self-learning, wisdom-generating AI machine became a reality?

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Accepted: 3 May 2023 / Published online: 6 June 2023
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I dedicate this editorial to the memory of Horst Rittel (1930–1990) who, in 1970, “proposed an Issue-Based Information System (IBIS) to support argument and discourse when dealing with wicked problems.” (Verma 2023, p. 4 of 7)

1 A self-learning, knowledge-generating AI machine that mastered board games

In his 1953 seminal paper “Digital computers applied to games,” the British mathematician and logician Alan Turing (1912–1954) writes with a vivid imagination of self-learning machine in mind, “Could one make a machine to play chess, and to improve its play, game by game, profiting from its experience? ... This could certainly be described as ‘learning,’ though it is not quite representative of learning as we know it.” (Turing 1953, p. 1 of 10, p. 10 of 10) Sixty-five years later, the first self-learning machine for board-game (including chess) play, AlphaZero, not only reached this milestone (McGrath et al. 2022, p. 1 of 10; Silver et al. 2018b) but also went above and beyond Turing’s imagination by virtue of its extraordinary “superhuman performance” (Campbell 2018; Silver et al. 2018a, p. 1140).

In a December 2018 editorial in the journal *Science*, the Russian chess grandmaster and former World Chess Champion Garry Kasparov marvels at AlphaZero, an artificial intelligence (AI) program—or “machine”, as Alan Turing would have called it—developed by the London-based British AI laboratory DeepMind, for its record-breaking performance in mastering board games. He commends, “AlphaZero is surpassing us (human board-game players and traditional AI board-game-playing programs) in a profound and useful way ...” (Kasparov 2018, p. 1087; the parenthesis by myself).

AlphaZero indeed merits such a high compliment. As a one-of-a-kind generalist program, it played chess, Go, and shogi (Japanese chess) at a level superior to not only top human players but also those traditional specialist programs that had previously won human–machine board-game competitions—Stockfish (for chess), AlphaGo (for Go), and Elmo (for shogi) (Campbell 2018; Kasparov 2018; Kissinger 2018, p. 12; Kissinger et al. 2019, pp. 1–2 of 7; McGrath et al. 2022, p. 1 of 10; Silver et al. 2018a, pp. 1140–1143; Silver et al. 2018b).

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Table 1 AlphaZero and proposed AlphaZero-*Ecophronesis*: their commonalities and differences

Attributes	AlphaZero (with proven attributes)	AlphaZero- <i>Ecophronesis</i> (? stands for “to be proven”)
Target problems	Tame problems of board games (see note 1 below)	Wicked problems in socio-ecological practice (see note 1 below)
Performance	“Superhuman,” extraordinary	“Superhuman,” extraordinary (?)
Strategies and actions as perceived by humans	Brilliant but bold, aggressive, counterintuitive, risky, even wrong	Brilliant but bold, aggressive, counterintuitive, risky, wrong, immoral (?)
Capabilities “to learn, improve, and surprise” (Kissinger et al. 2021, p. 4)	Self-learning, self-improving, acting solely on self-generated knowledge	Self-learning, self-improving, acting solely on self-generated wisdom (ecophronesis) (?)
Independence from human influence of any kind (except the basic game rules)	Yes	Yes (see note 2 below)
Benefits to humans	Machine knowledge—nonhuman knowledge generated without human biases—complementary to human knowledge (see note 3 below)	Machine ecophronesis—nonhuman ecological practical wisdom without human biases—complementary to human ecophronesis (?) (see note 4 below)

[1] Tame problems and wicked problems are discussed in Sect. 3. [2] Morality is a requisite for wisdom (Lekka-Kowalik 2021; Xiang 2023, pp. 3–5). Some claim that AI machines could be “morally better” than humans simply because they would learn to be moral without the influence of human biases (Lekka-Kowalik 2021, p.48) But could machines self-learn to be humanly moral or would they still need human input? [3] A recent research has found evidence for the existence of a broad set of humanly-understandable and useful concepts within the body of nonhuman knowledge AlphaZero generated (McGrath et al. 2022). [4] The potential benefits of machine ecophronesis are discussed in Sect. 4.

The developers and admirers attribute this *extraordinary performance* to the *extraordinary style* in which AlphaZero played: It is “a distinctive, unorthodox, yet creative and dynamic playing style” (Silver et al. 2018b) that has not been previously seen in board-game history (Campbell 2018; Kissinger 2018, p. 14; Kissinger et al. 2019, pp. 1–2 of 7; Silver et al. 2018b). Playing with this “most fascinating,” “amazing” style (Campbell 2018; Silver et al. 2018b), AlphaZero conceived and executed bold yet brilliant moves that neither follow any known game strategies nor conform to time-honored norms of human play; as such, in the eyes of human players, the moves it made often appeared to be counterintuitive, risky, or even wrong (Kasparov 2018; Kissinger et al. 2019, p. 2 of 7).

The extraordinary style stems from an *extraordinary dual capability* that AlphaZero possesses: the ability to acquire its own knowledge of game play without human assistance of any kind (except the basic game rules); and the ability to rely entirely on this self-generated knowledge in games against other players. Specifically, via a cut-and-try process called reinforcement learning, AlphaZero learns how to play a game from scratch by playing the game against itself with

no embedded human strategies but only the basic rules of the game (Campbell 2018; Kasparov 2018; McGrath et al. 2022, p. 1 of 10, p. 9 of 10; Silver et al. 2018a, p. 1140; Silver et al. 2018b). Once learned, it uses and *only uses* this body of self-acquired nonhuman knowledge to compete against other board-game players, whether they be humans or humanly-trained AI programs.

This extraordinary dual capability, according to the developers and admirers, confers on AlphaZero an *extraordinary competitive advantage* over traditional AI board-game-playing programs (e.g., Stockfish, AlphaGo, and Elmo) which rely largely if not entirely on human knowledge of game play as encoded by human programmers: That is the independence from human influence of any kind, including the priorities and prejudices of programmers (Kasparov 2018; McGrath et al. 2022, p. 1 of 10, p. 9 of 10; Silver et al. 2018a, p. 1140, p. 1144; Silver et al. 2018b). It is in the very sense of “having zero human knowledge in the loop,” as the chief designer David Silver explains, that the developers at DeepMind named it AlphaZero [see a video in Silver et al. (2018b) at <https://www.youtube.com/watch?v=7L2sUGcOgh0&t=101s>].

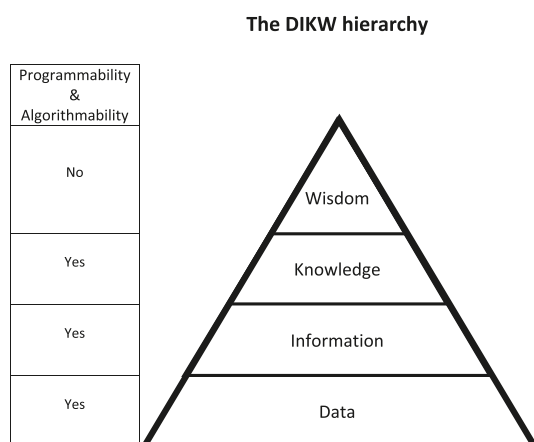


Fig. 1 The data–information–knowledge–wisdom (DIKW) hierarchy, the programmability and algorithmicability (i.e., being programmable and algorithmicable) of its four components (after Rowley 2007, p. 168). *Note:* [1] The conventional belief that wisdom is “non-algorithmic, non-programmable,” as shown here, would no longer be true if the self-learning, wisdom-generating AlphaZero-*Ecophronesis* with the attributes in Table 1 became a socio-technological reality. [2] All four elements in the DIKW hierarchy are necessary for humans to effectively undertake their various tasks in work and life, including the task of dealing with wicked problems in socio-ecological practice. However, being at the pinnacle of the hierarchy, wisdom plays an imperative role of paramount importance: It offers guidance, provides direction, and exercises control over data, information, and knowledge (Ackoff 1989, pp. 8–9; Lynch and Kaufman 2019, p. 455; Rowley 2007, p. 166). As a peculiar type of wisdom—the ecological practical wisdom, *ecophronesis* has the same chief authority over other three elements in the hierarchy; it decides what and how data, information, and knowledge should be used to effectively deal with wicked problems (a succinct definition of *ecophronesis* is given later in Sect. 3)

2 What if a self-learning, wisdom-generating AI machine became a reality?

While reading about this exemplary instance of human achievement and its fascinating progression from a twentieth century imagination to a twenty-first century socio-technological reality, I was intrigued by Garry Kasparov’s provocative conjecture that the human inventiveness leading to the AlphaZero wonder “may be duplicated on any other task or field where virtual knowledge can be generated” (Kasparov 2018, p. 1087).¹ My experience as a socio-ecological practice researcher gave me pause; my interest in how *ecophronesis* (ecological practical wisdom) enables

people to effectively undertake the task of dealing with wicked problems in socio-ecological practice aroused my curiosity²:

Now that we humans can make and use AlphaZero that self-learns to be superintelligent in mastering board games, IF we could also make AlphaZero-*Ecophronesis*, a sibling AI program or machine that, as the name implies, could self-learn to be *ecophronetic*—socio-ecologically and practically wise—in dealing with wicked problems (see Table 1), WHAT would the hitherto human task of dealing with wicked problems in socio-ecological practice look like?

Admittedly, the question, by virtue of its disruptive premise, would necessarily invite bold even wild imaginations beyond the present scope of human reason. As I shall show in the next section, the premise that a self-learning, wisdom-generating AI machine AlphaZero-*Ecophronesis* became a socio-technological reality, following in the footsteps of its knowledge-generating sibling AlphaZero, is profoundly disruptive at least on several important counts.

3 How profoundly disruptive is the premise?

First, it means that the conventional belief that wisdom is “non-algorithmic, non-programmable” (Rowley 2007, p. 168, citing Awad and Ghaziri 2004) were no longer true; and that AlphaZero-*Ecophronesis* as a self-learning, wisdom-generating machine could operate *independently* at the top level of the classic data–information–knowledge–wisdom (DIKW) hierarchy (Fig. 1). Specifically, with a dual capability comparable to that of AlphaZero (see Sect. 1 and Table 1), not only could AlphaZero-*Ecophronesis* generate its own ecological practical wisdom—*machine ecophronesis*, but it could also rely entirely on it in dealing with wicked problems; and it could conduct these learning and acting activities entirely on its own in a virtual world.

The second aspect of the premise’s disruptiveness is a logical yet fatal extension of the first. In his presidential address to the 1988 annual meeting of the International Society for General Systems Research (ISGSR) in St. Louis,

¹ Garry Kasparov is not alone nor the first. For example, 2 years after Alan Turing presented his seminar idea (Turing 1953), the American computer scientist John McCarthy (1927–2011) expressed the hope “to program a machine to learn to play (board) games well *and do other tasks*.” (McCarthy 1955; the parenthesis and italics by myself).

² [1] The concepts of *ecophronesis* and wicked problems will be discussed in Sect. 3. [2] “Socio-ecological practice is the human action and social process that take place in specific socio-ecological context to bring about a secure, harmonious, and sustainable socio-ecological condition serving human beings need for survival, development, and flourishing. It is the most fundamental and arguably primordial social practice Homo sapiens has been involuntarily engaging in over thousands of years of co-evolution with nature. Socio-ecological practice includes six distinct yet intertwining classes of human action and social process—planning, design, construction, restoration, conservation, and management.” (Xiang 2019, p.7)

Table 2 The ten attributes of wicked problems (Chan and Xiang 2022, p. 2)

1	There is no definitive formulation of a wicked problem
2	Wicked problems have no stopping rule
3	Solutions to wicked problems are not true-or-false, but good-or-bad
4	There is no immediate and no ultimate test of a solution to a wicked problem
5	Solutions to wicked problems trigger ripple effects that are neither reversible, nor stoppable
6	Wicked problems have no clear solution, and perhaps not even a set of possible solutions
7	Every wicked problem is essentially unique
8	Every wicked problem can be a symptom of another problem
9	Every wicked problem can be explained in numerous ways. The choice of explanation determines the nature of the problem's resolution
10	The planner (designer and policymaker) has no right to be wrong

[1] Wicked problems are a class of infamous social system (including socio-ecological system where socio-ecological practice takes place) problems. Their inherently “aggressive,” “malignant,” and “vicious” attributes as summarized here along with their daunting ubiquity and notorious eternity make them “mischievous” and “evil” natural enemies of the human race (Churchman 1967, p.B-141; Rittel and Webber 1973, p. 160; Xiang 2013, p. 2; Xiang 2019, p. 8). Wicked problems are present in essentially all issue areas of socio-ecological practice that matter to the human society today, ranging from global climate change adaptation and mitigation, human sustainability, urbanization, nature-based solutions (NBS), to equity, poverty, hunger, and the common threat posed by pandemics (Chan 2014, pp. 18–19; Chan and Xiang 2022, p. 2; La Rosa et al. 2021, pp. 331–332; Verma 2023, p. 5 of 7; Xiang 2013, p. 2; 2019, p.8; 2021a, p. 77; 2021b, pp. 239–240). [2] In the table, descriptions of attributes 1, 2, 3, 4, and 7 are the exact quotes from Rittel and Webber (1973); descriptions of other attributes are either paraphrases or edited quotes of Rittel and Webber (1973) by Buchanan (1992, p. 16), Peters (2017, p. 388), and Xiang (2013, p. 1)

Missouri, USA, the American management science scholar Russell Ackoff (1919–2009) famously enunciates:

“... wisdom-generating systems are ones that man will never be able to assign to automata. It may well be that wisdom, which is essential to the effective pursuit of ideals, and the pursuit of ideals itself, are the characteristics that differentiate man from machines.” (Ackoff 1989, p. 9)³

Obviously, this distinction between humans and machines would unequivocally become blurred if not simply disappeared should AlphaZero-*Ecophronesis* with attributes tabulated in Table 1 become a reality.

Third, the premise means that the existent conceptions of wisdom, including those of ecophronesis, were no longer accurate. Among other things, wisdom would not be a virtue belonging solely to humans anymore, nor would there be only a human way to learn and exercise wisdom. A case in point is a recent conception of ecophronesis—the virtue of ecological practical wisdom. According to Xiang (2023, p. 5), ecophronesis is “the human ability par excellence to make

and implement morally sound and pragmatically effective choices in the complex, heterogeneous situations of socio-ecological practice”; and people learn to be ecophronetic by simply doing ecophronetic things in professional and/or academic practices, including moral improvisation, reflection on experience, emulation of ecophronimoi—people of ecophronesis (plural form for ecophronimos), and journal writing (*Ibid.*, p. 3, pp. 6–7).⁴ Clearly, this conception would need to be revisited and augmented if the premise were true.

Fourth, the premise means that the problems AI programs or machines can tackle were no longer limited to what Horst Rittel (1930–1990), Melvin Webber (1920–2006), and West Churchman (1913–2004) dub “tame problems” (Churchman 1967; Rittel and Webber 1973).⁵ Tame problems possess a set of attributes that arguably make themselves an easy and fitting target of conventional AI programs. These attributes include, but are not limited to, (1) being well-defined, routine, repetitive; (2) having clear goals and following rigid rules; (3) taking place in a closed, rather than an open,

³ As one of the pioneers in operations research, systems thinking, and management science, Russell Ackoff was actively involved in ISGSR. The society began in the early 1950s as the Society for the Advancement of General Systems Theory, changed its name to the Society for General Systems Research in the fall of 1955. The name was changed again in 1986 to the International Society for General Systems Research (ISGSR), and then finally, in 1988, to the International Society for the Systems Sciences (ISSS, <https://www.issss.org>) [<https://www.issss.org/history/> (accessed March 30, 2023)].

⁴ For various conceptions of ecophronesis, see Austin (2018), Lu and Wang (2022, in Chinese), and Xiang (2016; 2023, pp.3–5); for the usefulness of ecophronesis to socio-ecological practice, see Achal and Mukherjee (2019), Grose et al. (2019), Heavers (2023), Jiang et al. (2022), Li et al. (2021), Wang (2019), and Yang and Young (2019), among others.

⁵ All three were erstwhile faculty members at the University of California at Berkeley, USA. On the contributions of their wicked-problems work to planning and design, more recent publications include Chan (2023), Chan and Xiang (2022), and Verma (2023); and on contributions to public policy, Head (2022).

system; (4) being decomposable; and (5) having optimal or ultimate solutions (De Cremer and Kasparov 2021, p. 3 of 8; McCarthy et al. 1955/2006, p. 14). Evidently, none of these attributes is found in wicked problems (Table 2). As such, AlphaZero-*Ecophrones*' ability to deal with wicked problems on its own would be truly ground-breaking should it be materialized.

4 A human-machine-partnership scenario

What would the hitherto human task of dealing with wicked problems in socio-ecological practice look like if AlphaZero-*Ecophrones* with attributes tabulated in Table 1 became a socio-technological reality? This question, with its disruptive if-clause above-discussed, invites us socio-ecological practice researchers [aka ecopracticologists, Xiang (2019, p. 12)] to envision alternative futures that are deemed surprising and uneasy. But at the same time, responding to the question and questions of this kind helps emancipate us from confinement and restraint of conventional thinking that we have long been taking for granted and feeling comfortable with (Kissinger et al. 2021, pp. 206–207, pp. 211–213). As such, the question warrants our due responses. In this very spirit, I dare to share my imaginations via a scenario in the hope that others will join in this liberating and meaningful intellectual exercise to “stretch and focus” our thinking on alternative futures (Xiang and Clarke 2003, p. 899).

Under this human-machine-partnership scenario,

1. In socio-ecological practice where wickedness—the ubiquity of wicked problems—is a daunting reality (see note [1] in Table 2), practitioners team up with AlphaZero-*Ecophrones* to undertake the task of dealing with wicked problems cooperatively. They do so with the belief that dealing with wicked problems should not be completely outsourced to an AI machine—however competent it may be, but the task will benefit greatly from a collaborative human-machine partnership.
2. On this human-machine team, practitioners are willing and able to first learn about *machine ecophrones* AlphaZero-*Ecophrones* generates (Table 1) and then form an *augmented ecophrones* by mingling machine ecophrones with their own *human ecophrones*. In so doing, they follow in the footsteps of those human board-game players who, as De Cremer and Kasparov (2021), McGrath et al. (2022), and Silver et al. (2018b) point out, improved their own play by learning about AlphaZero's machine intelligence and emulating its playing style. The creation of an augmented ecophrones is inspired by the belief of many AlphaZero's admirers that an augmented intelligence, which combines the strengths of human intelligence with those

of machine intelligence, provides an opportunity for humans to advance to higher levels of understanding, expansive thoughts, and self-awareness (e.g., De Cremer and Kasparov 2021, pp. 4–8 of 8; Hultin 2019; Johnson and Sivas 2018; Kasparov 2018; Kissinger et al. 2019, p. 1 of 7, p. 5 of 7; Kissinger et al. 2021, pp. 206–207, pp. 211–218).⁶

3. The augmented ecophrones combines strengths of machine ecophrones with those of human ecophrones and avoids many human weaknesses in judgment-making, including systematic cognitive biases embedded in heuristics, such as those identified collectively by the Israeli psychologist Amos Tversky (1937–1996) and the Israeli–American psychologist Daniel Kahneman in representativeness heuristic, availability heuristic, and adjustment heuristic (Tversky and Kahneman 1974)⁷; as such, the augmented ecophrones allows practitioners to transcend ordinary limits for human cognition and wisdom, and advance to an extraordinary level of human wisdom that is neither imaginable nor achievable with human ecophrones alone.
4. With the augmented ecophrones, practitioners undertake the task of dealing with wicked problems in a way that has not been previously seen in the history of socio-ecological practice. Among things that we can imagine at present, they prudently transform certain wicked

⁶ Although wisdom and intelligence are widely regarded as distinct yet related human capabilities (Lynch and Kaufman 2019; Song 2021; Staudinger and Glück 2011), the great majority of AI literature draws no distinction between them. Among the few exceptions are two recent articles tellingly entitled “From artificial intelligence to artificial wisdom: what Socrates teaches us” (Kim and Mejia 2019) and “Morality in the AI world” (Lekka-Kowalik 2021), respectively. In the latter article, the author exclaims in delight that “AI ethics finally aims at artificial wisdom, or at wise artificial intelligence.” (Lekka-Kowalik 2021, p.48).

⁷ [1] The use of heuristics is a double-edged sword. “Evidently, people have a natural tendency to conduct their cognitive activities under ease-based heuristics. These are cognitive procedures or judgmental strategies that are simple, easy, and useful, on the one hand (Nisbett and Ross 1980, pp.254–255), but are narrow, shallow, and often biased or even misleading, on the other (Bazerman 2002, pp.11–40; Heath et al. 1998, p.2; Nisbett and Ross 1980, pp.17–23, pp.41–42; Tversky and Kahneman 1974, pp.1130–1131).” (Xiang and Clarke 2003, p.889) [2] According to the Polish philosopher Agnieszka Lekka-Kowalik, machine practical wisdom (i.e., machine ecophrones in our case) which she calls “artificial wisdom” (Lekka-Kowalik 2021, p. 44, pp. 47–48) has some distinctive advantages over human practical wisdom (i.e., human ecophrones in our case). She writes, “Some thinkers even claim that AI would be ‘morally better’ than human beings, for human moral judgments are disturbed by emotions, partiality, and individual vices. AI is immune to such disturbances. It is also better at collecting data, analyzing facts, discovering alternatives, and so on, so its judgments should be better. Moreover, AI may face moral dilemmas more often; there could be moral dilemmas that humans have never faced because of our limited capacities. So, it seems that human experience would not be sufficient for equipping AI with morality.” (Lekka-Kowalik 2021, p.48).

problems into tame problems by eliminating all the ten attributes (Table 2); convert some other wicked problems into less-wicked ones by removing or mitigating some of the ten attributes; and develop novel strategies for working with remaining wicked problems as well as those less-wicked ones.

5. The human–machine team finds itself engaged in a continuously repeated learning–acting cycle as new mutations of wicked problems emerge. Through such a process, both practitioners and AlphaZero-*Ecophrones* improve themselves in a coherent way toward a mutually beneficial end—better socio-ecological practice; more productive and fulfilled practitioners; and a more effective and humble AI machine.
6. Still, there is a less tangible yet more profound achievement: the emergence and progression of a harmonious human-machine relationship between practitioners and AlphaZero-*Ecophrones*. In the eyes of practitioners, AlphaZero-*Ecophrones* has proved itself neither a fancy “marvelous toy”, as Klaus Barber and Susan Rodman once dubbed FORPLAN—a short-lived computerized optimization model for forest planning and management at the US Forest Service in the 1980s (Barber and Rodman 1990), nor a competitor that might replace practitioners in the socio-ecological practice of dealing with wicked problems. Instead, it is a serious, useful, independent yet cooperative partner, and “a fellow ‘being’ experiencing and knowing the world—a combination of tool, pet, and mind.” (Kissinger et al. 2021, p. 212)

It is noteworthy that the antithesis of the human-machine-partnership scenario is a human-machine-rivalry scenario. Under that scenario, neither humans nor AI machines can conceive of themselves as being in a positive-sum game where cooperations lead to mutual gains (as in the human-machine-partnership scenario); instead of partners, they choose to be competitors fighting with each other in a zero-sum or even a negative-sum game for hegemony [for a succinct description about these three game situations in game theory, see The Editors of Encyclopaedia Britannica (2015)]. As I shall show below with a recent example, the rivalry-scenario thinking undergirds many contemporary AI apprehensions and criticisms.

5 Embracing opportunities and challenges of the ever-evolving human inventiveness

In a March 22, 2023 open letter signed by thousands of people from around the world, Future of Life Institute (<https://futureoflife.org/>) calls on all AI laboratories “to immediately pause for at least 6 months the training of AI systems more

powerful than GPT-4 (an AI program developed by the San Francisco, California-based American AI lab OpenAI).” (Future of Life Institute 2023, the parenthesis by myself) It urges AI developers to use this pause working with policy-makers “to dramatically accelerate development of robust AI governance systems” in the hope that such governance systems help mitigate or avoid “potentially catastrophic effects (of AI programs or machines) on society” (*Ibid.*, the parenthesis by myself).

According to the open letter, the call reflected a deep concern at a time when “AI labs locked in an out-of-control race to develop and deploy ever more powerful digital minds that no one—not even their creators—can understand, predict, or reliably control.” (Future of Life Institute 2023) The concern was that AI programs or machines that are no longer constrained by the limits of human knowledge or intelligence are “becoming human-competitive at general tasks” so much so that one day, they “might eventually outnumber, outsmart, obsolete, and replace us (humans)” (*Ibid.*, the parenthesis by myself). A terminator-like apocalypse seems on the horizon under this apparent human-machine-rivalry-scenario thinking.

While the open letter highlights the paramount need for a socio-technological ethic to comprehend and even guide the AI development and use, it raises a broader and more fundamental question: How should we human beings respond to the opportunities and challenges of our own ever-evolving inventiveness? Throughout human history, the question has arisen multiple times in response to seismic innovations that promised to both enrich and disrupt established ways of work and life. These include, but are not limited to, inventions of papermaking, printing, gunpowder, the compass, the steam engine, the Internet, and arguably the atomic bomb. This time however the question concerns an unprecedented disruptive socio-technological revolution which may lead to a world relying heavily on self-learning AI machines rarely if ever constrained by human knowledge, intelligence, and wisdom.⁸

For us members of the socio-ecological practice research community (the SEPR community), there are two pressing pertinent issues:

As self-learning AI machines become ever more sophisticated, more knowledgeable, and even wiser beyond the present scope of human reason and imagination, they could increasingly undertake many tasks in socio-ecological practice hitherto undertaken solely

⁸ For more systematic explorations of challenges as well as opportunities AI brings to human beings, see Kissinger et al. (2021) and Russell (2019); critical examinations can also be found in, among others, Bender et al. (2021), Berendt (2018), Brundage et al. (2018), Candelon et al. (2021), Cellan-Jones (2014), Chisnall (2020), Hammershøj (2019), Kissinger (2018), Koch (2019), Manyika et al. (2017), Oluwaniyi (2023), Ossewaarde (2019), and Ossewaarde and Gülenç (2020).

by humans, including potentially the task of dealing with wicked problems; they could also fulfill these tasks in their own way—nonhuman way, that is—as aforementioned AlphaZero did in mastering board games. What opportunities and challenges would this socio-technological revolution present to us? How should we respond?

Since the ongoing AI revolution is driven by the endless, restless human inventiveness and deemed neither stoppable nor reversible (AI for Good 2023; Floridi and Cows 2019; HAI 2022; Kissinger et al. 2021; Leeming 2021; Tomašev et al. 2020; U4SSC 2017, 2020, 2021), forgoing questions of this kind, including more specific ones—like the what-if question in Sect. 2, is simply infeasible. I, therefore, developed this essay within the confines of my own knowledge, experience, and imagination in the hope that it serves as a starting point or a springboard for an open, explorative dialogue among members of the SEPR community and other academic and professional communities. In the optimistic spirit that the future will favor those members of humanity who choose to courageously and prudently embrace both opportunities and challenges of the ever-evolving human inventiveness, let's act proactively: asking questions about implications of the AI revolution to socio-ecological practice; liberating ourselves from the restraints of present knowledge, experience, and wisdom; and cooperating and coevolving with this new and powerful player in our noble socio-ecological endeavor to build a better world.

Acknowledgements I thank the following individuals for their help during the preparation of this editorial (in alphabetic order): Cristina Bueti (International Telecommunication Union, Geneva, Switzerland), Taylor Jones (Future of Life Institute, Cambridge, Massachusetts, USA), David Silver (DeepMind, London, UK), Wenwu Tang and Tianyang Chen (University of North Carolina at Charlotte, USA). My interest in artificial intelligence began in the 1980s when it was commonly called “expert systems” or “knowledge-based systems.” Over the past four decades, especially in the last 10 years, I witnessed tremendous growth in the field. I thank the following individuals who introduced me to the field: Julius Gyula Fábos, Bruce MacDougall, Meir Gross (University of Massachusetts at Amherst, USA), and Hellen Couclelis, in particular, her 1986 article (Couclelis 1986) (University of California at Santa Barbara, USA).

Author contributions The author was fully responsible for the conception and writing of the editorial.

Funding The writing of this editorial was not supported by any funding.

Declarations

Conflict of interest The author confirms that there is no conflict of interest.

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